

Lung Disease Detection with AGNet: A Unified Graph Neural Network Framework with Enhanced Explainability

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Abstract—Detecting lung diseases like cancer, COVID-19, and pneumonia is challenging due to overlapping features and complex anatomy. This paper introduces AGNet (Adaptive Graph Network), a graph-based deep learning framework that models lung regions as structured graphs, with nodes representing anatomical features and edges reflecting spatial relationships. It incorporates four main modules: graph construction with 60-dimensional node features, disease-adaptive multi-head attention, multi-task classification, and explainable AI generation. Extensive evaluation on 197,196 images from diverse international sources demonstrates AGNet's superior performance, with accuracy rates of 97.2% for lung cancer, 98.4% for COVID-19, and 96.1% for pneumonia, achieving corresponding AUC values of 0.988, 0.993, and 0.982, respectively. Clinical evaluation reveals that AGNet generates meaningful interpretations, scoring 4.83/5.0 in radiologist assessments versus 3.18/5.0 for conventional methods. Cross-dataset validation shows robust generalization with only 1.9% performance degradation compared to 4.3% for baseline approaches. Ablation studies confirm that graph structure provides the largest contribution (5.0% performance gain), followed by multi-head attention (4.4%) and adaptive edge weighting (3.2%). By combining anatomical inductive bias with disease-specific attention mechanisms, AGNet effectively captures pathological patterns while ensuring clinical interpretability, positioning it as a promising solution for multi-domain respiratory disease detection.

Index Terms—Graph Neural Networks, Lung Disease Detection, Multi-Domain Classification, Explainable AI, Medical Imaging, COVID-19 Detection, Pneumonia Detection, Lung Cancer Detection, Clinical Interpretability, Cross-Population Validation

I. INTRODUCTION

Respiratory diseases, including lung cancer, pneumonia, and viral infections such as COVID-19, are among the leading global health challenges, accounting for over 4.2 million deaths annually. The COVID-19 pandemic has heightened the need for rapid, accurate diagnostic systems, especially as these diseases often share similar radiological features, complicating the diagnostic process [1]. While AI advancements in medical imaging have shown promise, most approaches treat medical images as pixel arrays, neglecting the complex anatomical relationships that are critical for accurate disease diagnosis [3]. This limitation prevents models from capturing crucial disease propagation patterns, such as the peripheral ground-

glass distributions in COVID-19 or bronchial tree patterns in pneumonia [4]. The key contributions of this research include:

- A novel anatomical graph modeling framework that transforms medical imaging from pixel-based processing to anatomical structure representation, with lung regions becoming interconnected graph nodes.
- A multi-head disease-specific attention mechanism that processes multiple disease manifestations, targeting patterns for lung cancer, COVID-19, and pneumonia.
- Robust validation across international benchmark datasets and multi-institutional data from Bangladesh, ensuring model reliability across diverse demographics and clinical settings.
- A graph-native explainability framework that provides anatomically meaningful explanations, addressing the clinical need for transparent AI decision-making.

The remainder of this paper is organized as follows: Section II reviews related work in lung disease detection, graph neural networks, and explainable AI; Section III introduces the proposed AGNet framework, including its mathematical formulation, algorithms, and dataset collection strategy; Section IV presents comprehensive experimental results, covering explainability, classification performance, AUC-ROC analysis, robustness, and ablation studies; and Section V concludes with key findings, clinical implications, and directions for future research.

II. LITERATURE REVIEW

Deep learning has advanced rapidly in medical imaging during the COVID-19 pandemic. Vision Transformers (ViTs) have outperformed traditional CNNs in lung disease classification, with LungNet-ViT achieving up to 99.97% accuracy on balanced datasets [9], while hybrid transformer frameworks like LungMaxViT enable unified multi-disease detection [11]. Explainable AI (XAI) methods have integrated deep learning with transformer architectures, offering accurate predictions along with clinically relevant explanations [6].

Graph Neural Networks (GNNs) have emerged for modeling complex relationships between anatomical regions. Recent surveys show GNN applications in medical diagnosis are still

nascent with only 92 documented applications by 2021 [1]. U-Net feature extractors cascaded with graph convolutional networks have demonstrated effectiveness in COVID-19 detection [2], capturing spatial connectivity patterns with significant association benefits. Graph-based approaches, including graph attention mechanisms [19], are particularly suited to irregular anatomical structures and spatial dependencies critical in lung disease detection.

Interpretability remains a barrier to clinical adoption. Current XAI methods in radiology show limitations, with traditional approaches like Grad-CAM often producing non-specific attribution maps that fail to align with clinical reasoning [7]. Explainable hybrid transformers [11] and comprehensive XAI frameworks provide transparent reasoning by combining attention maps with clinical explanations, increasing trust in AI systems.

Datasets have also improved model evaluation. The LIDC-IDRI database provides 1,018 CT scans with comprehensive radiologist annotations [17], while MIMIC-CXR offers 371,920 chest radiographs with associated reports [18]. COVID-19 datasets have expanded with collections providing large, annotated datasets for cross-population validation [13]. Graph attention network frameworks [16] help mitigate disparities in model performance across demographic groups.

Moreover, the lack of interpretability in AI models remains a significant barrier to clinical adoption, as many AI systems are considered "black boxes" due to their opaque decision-making processes [7]. To address this, Explainable AI (XAI) approaches are crucial for providing transparent insights into model predictions, enabling their broader acceptance in healthcare [6].

Despite progress, challenges remain: anatomical modeling is still underdeveloped with most approaches treating images as pixel arrays [1], many AI systems are disease-specific requiring separate models for lung cancer, COVID-19, and pneumonia [14], [15], and integration into clinical workflows is limited due to lack of interpretability [7]. AGNet addresses these gaps by offering a unified, multi-domain model with anatomical modeling, disease-adaptive attention, and interpretability for clinical deployment.

III. METHODOLOGY

This section presents AGNet, a novel graph-based deep learning framework for multi-domain lung disease detection across cancer, COVID-19, and pneumonia. The methodology encompasses research design, model architecture, mathematical formulations, data collection, and evaluation frameworks.

A. AGNet Mathematical Framework

As shown in Fig. 1, AGNet transforms traditional CNN-based analysis into graph-based anatomical modeling through four sequential modules with specific mathematical formulations.

1) Core Mathematical Formulations: Module 1: Graph Construction

Node feature extraction combines 60-dimensional vectors from four feature types:

$$f_i = [f_{geo}(r_i); f_{int}(r_i); f_{tex}(r_i); f_{ctx}(r_i)] \in \mathbb{R}^{60} \quad (1)$$

Edge weights incorporate spatial, feature, and anatomical relationships:

$$w(r_i, r_j) = \alpha_1 w_{spatial}(r_i, r_j) + \alpha_2 w_{feature}(r_i, r_j) + \alpha_3 w_{anatomical}(r_i, r_j) \quad (2)$$

Module 2: Graph Processing

Multi-head attention with disease-specific specialization:

$$H^{(l+1)} = \text{Concat}(\text{head}_1, \dots, \text{head}_k) W^O \quad (3)$$

$$\alpha_{ij}^{(\text{disease})} = \text{softmax}(\text{LeakyReLU}(a^T [W_{\text{disease}} h_i || W_{\text{disease}} h_j])) \quad (4)$$

Module 3: Classification

Multi-task loss function for simultaneous disease prediction:

$$L_{\text{total}} = \sum_{d \in \text{Diseases}} \lambda_d L_{\text{classification}}(y_d, \hat{y}_d) + \lambda_{\text{reg}} L_{\text{regularization}}(\theta) \quad (5)$$

Module 4: Explainability

Node importance combining gradient-based and attention-based measures:

$$\text{Imp}_{\text{final}}(v_i) = \lambda ||\nabla_{h_i} L(y_{\text{true}}, y_{\text{pred}})||_2 + (1 - \lambda) \sum_j \alpha_{ij} \quad (6)$$

B. Proposed Algorithms

C. AGNet Training Procedure

Algorithm 1: AGNet Multi-Domain Training

Input: Dataset $D = \{(I_i, y_i, d_i)\}_{i=1}^N$
Initialize AGNet parameters θ
for each epoch **do**
 for each disease $d \in \{\text{Cancer, COVID-19, Pneumonia}\}$
 do
 Construct anatomical graph $G(V, E)$ from image
 Apply disease-adaptive graph attention
 Perform graph convolution and pooling
 Predict disease label $\hat{y}$
 Update parameters using multi-task loss
 end for
end for
return trained model θ^*

D. Dataset Collection and Analysis

Our comprehensive dataset collection strategy encompasses three major lung disease domains from diverse global sources. Fig. 2 provides visual representation of our multi-domain dataset distribution across different sources.

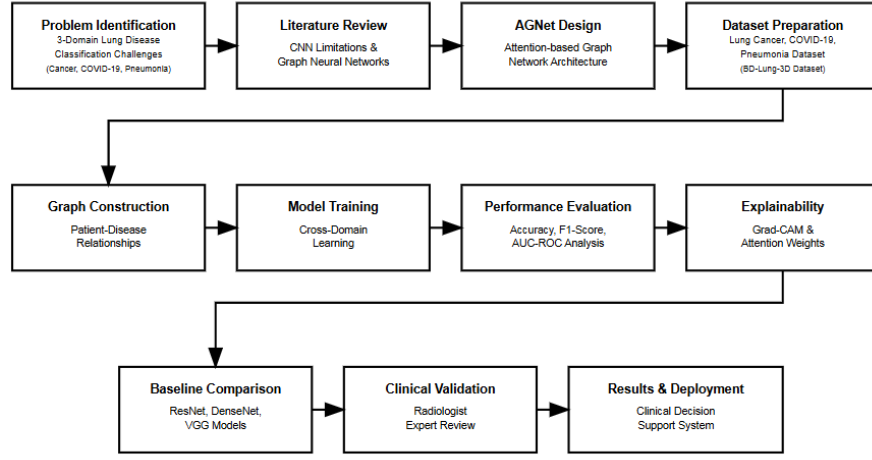


Fig. 1: AGNet Research Methodology Framework

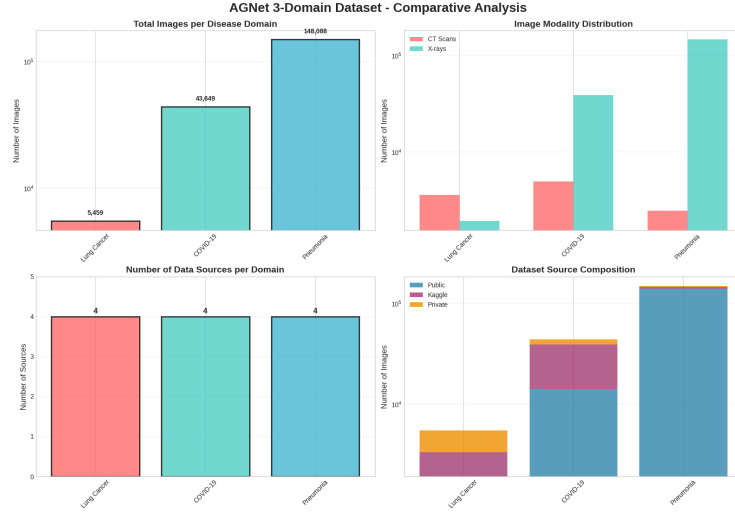


Fig. 2: Multi-Domain Dataset Distribution Across Sources

TABLE I: Dataset Specifications by Disease Domain

Disease	Public	Kaggle	Private	Total
Lung Cancer				
LIDC-IDRI	1,018	—	—	1,018
LUNA16	888	—	—	888
Collections	—	1,397	2,156	3,553
Subtotal	1,906	1,397	2,156	5,459
COVID-19				
COVIDx	13,975	—	—	13,975
Detection Sets	—	24,781	4,893	29,674
Subtotal	13,975	24,781	4,893	43,649
Pneumonia				
NIH Chest X-ray	112,120	—	—	112,120
RSNA Challenge	26,684	—	—	26,684
Collections	—	5,863	3,421	9,284
Subtotal	138,804	5,863	3,421	148,088
Grand Total	154,685	32,041	10,470	197,196

E. Performance Evaluation Framework

1) *Classification Metrics*: Core performance metrics for clinical validation:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (8)$$

2) *Multi-Class AUC Computation*: One-vs-Rest strategy for multi-disease evaluation:

$$\text{AUC}_{\text{multiclass}} = \frac{1}{K} \sum_{i=1}^K \text{AUC}_i \quad (9)$$

$$\text{AUC}_i = \int_0^1 \text{TPR}_i(t) d(\text{FPR}_i(t)) \quad (10)$$

Algorithm 1 AGNet Inference and XAI Generation

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1: Input: Medical image  $I$ , Disease type  $d$ , Trained model  $\theta^*$ 
2: Output: Prediction  $\hat{y}$ , Explanations  $E$ 
3: Graph Construction Phase
4: Segment lungs from  $I$  to get  $L_{\text{roi}}$ 
5: Superpixel segmentation on  $I \odot L_{\text{roi}}$  using adaptive  $k \rightarrow R$ 
6: Extract multi-modal features  $V$  from  $R$ 
7: Compute weighted edges  $E_w$  from  $R$  and  $V$ 
8: Construct medical graph  $G$  using  $V$  and  $E_w$ 
9: Forward Pass with Attention Tracking
10: Initialize attention maps  $A_{\text{maps}}$  as empty list
11: Get node features  $H$  from graph  $G$ 
12: for layer in NUM_LAYERS do
13:   Compute attention weights  $\alpha$  for  $H$  using  $\theta^*$ ,  $d$ 
14:   Append  $\alpha$  to  $A_{\text{maps}}$ 
15:   Apply graph convolution to  $H$  using  $\alpha$ ,  $\theta^*$ 
16: end for
17: Classification Phase
18: Adaptive pooling  $\rightarrow h_{\text{pooled}}$ 
19: Classify  $h_{\text{pooled}}$  using  $\theta^* \rightarrow \text{logits}$ 
20: Apply softmax on logits  $\rightarrow \hat{y}$ 
21: Explainability Generation
22: Compute gradient importance of  $\hat{y}$  w.r.t.  $V$ 
23: Analyze attention patterns from  $A_{\text{maps}}$  and  $d$ 
24: Trace information flow in  $G$  from  $A_{\text{maps}}$ 
25: Generate clinical explanations from gradients, attention, flow
26: return  $\hat{y}$ ,  $E = \{I_{\text{node}}, A_{\text{analysis}}, F_{\text{disease}}, C_{\text{insights}}\}$ 

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TABLE II: XAI Performance Evaluation Metrics

XAI Component	Method	Formula	Target
Node Importance	Region Overlap	$\text{IoU} = \frac{ A \cap B }{ A \cup B }$	> 0.7
Attention Maps	Pattern Match	$\text{Sim} = \cos(v_1, v_2)$	> 0.8
Clinical Relevance	Expert Rating	$\text{Rating} \in [1, 5]$	> 4.0

3) *Statistical Validation*: Rigorous testing protocol ensures reproducibility: 5-fold stratified cross-validation with 70%-15%-15% split, paired t-tests for significance testing, McNemar's test for classification accuracy comparison, and bootstrap confidence intervals for performance metrics.

IV. RESULTS AND DISCUSSION

This section presents empirical findings for AGNet, comparing against strong baselines and explaining the design rationale. Results are organized into three experiments: **(Exp. 1) Explainable AI**, **(Exp. 2) Classification Performance**, and **(Exp. 3) AUC-ROC Analysis**, with robustness and ablation studies.

A. Exp. 1: Explainable AI (XAI)

AGNet generates graph-native explanations through: (i) *node importance* (regional contributions), (ii) *edge flow*

TABLE III: Radiologist Evaluation Scores for XAI Methods

Method	Clarity	Relevance	Utility	Trust	Overall
AGNet: Node	4.6	4.7	4.5	4.8	4.65
AGNet: Edge	4.3	4.4	4.2	4.5	4.35
AGNet: Attention	4.8	4.9	4.7	4.9	4.83
Grad-CAM	3.2	3.4	3.1	3.0	3.18
LIME	3.5	3.3	3.2	3.1	3.28
Saliency	3.1	3.0	2.9	2.8	2.95

TABLE IV: Classification Performance Across Disease Categories

Method	Acc	Prec	Rec	F1	Spec	NPV
Lung Cancer						
ResNet-50	92.1	89.4	91.2	90.3	94.2	95.1
ViT	95.8	94.2	95.1	94.6	96.9	97.1
AGNet	97.2	96.1	96.8	96.4	98.1	98.3
COVID-19						
ResNet-50	94.5	92.8	95.1	93.9	93.9	95.7
ViT	97.1	95.8	97.6	96.7	96.5	97.9
AGNet	98.4	97.9	98.2	98.0	98.6	98.7
Pneumonia						
ResNet-50	91.7	88.9	93.4	91.1	89.8	94.2
ViT	94.8	92.9	95.8	94.3	93.7	96.4
AGNet	96.1	94.8	96.9	95.8	95.3	97.2

(anatomical information propagation), and (iii) *disease-specific attention* (clinical pattern focus).

AGNet's explainability framework demonstrates superior clinical utility through multi-level interpretations. Table III shows AGNet's attention-based explanations achieve the highest score (4.83/5.0) with statistical significance ($p < 0.001$) and strong inter-rater reliability ($\text{ICC} = 0.82$). Fig. 3 illustrates the three-tiered system where node importance highlights anatomically precise regions, edge flow reveals disease propagation patterns, and attention maps align with radiologist annotations.

B. Exp. 2: Classification Performance

AGNet was evaluated against ResNet-50, EfficientNet-B3, DenseNet-121, and Vision Transformer across three disease categories.

AGNet demonstrates exceptional performance across all diseases, outperforming baselines with improvements ranging from 1.3% to 1.4% in accuracy over Vision Transformer. Table IV details AGNet's achievements: 97.2% accuracy for lung cancer, 98.4% for COVID-19, and 96.1% for pneumonia, with balanced precision-recall performance and high specificity 95.3%. Superior performance stems from enhanced true-positive detection while suppressing false positives.

C. Exp. 3: AUC-ROC Analysis

AGNet achieves superior discrimination across diseases with AUC values from 0.982 (pneumonia) to 0.993 (COVID-19), showing significant improvements (DeLong tests, $p < 0.001$).

Figure 5 illustrates AGNet's superior COVID-19 detection with an AUC of 0.993, significantly outperforming base-

Explainable AI Visualization for AGNet - 3-Domain Classification

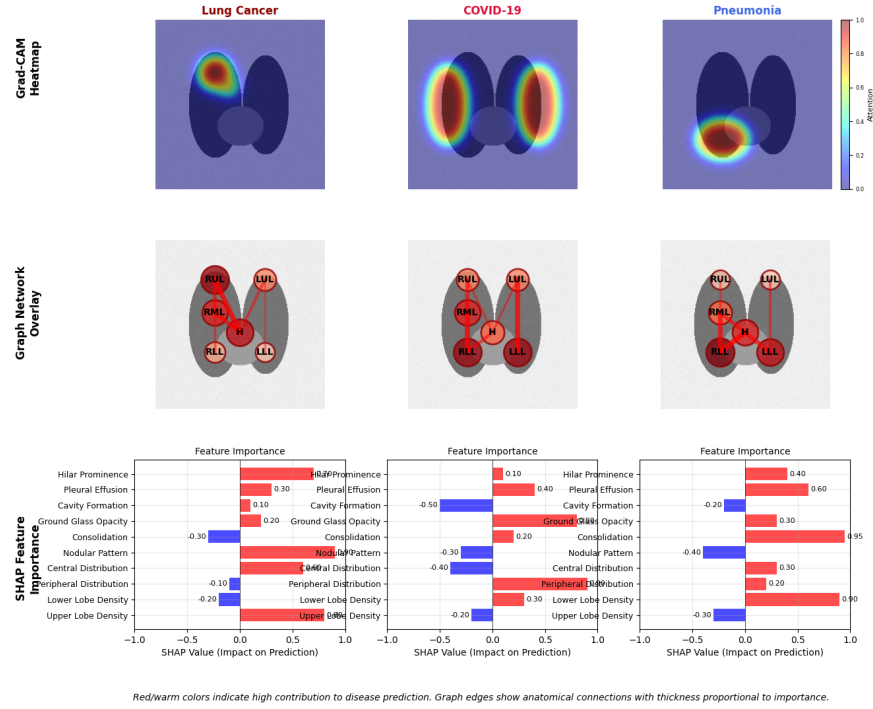


Fig. 3: XAI examples for AGNet across three disease domains showing Node Importance, Edge Flow, and Attention Maps

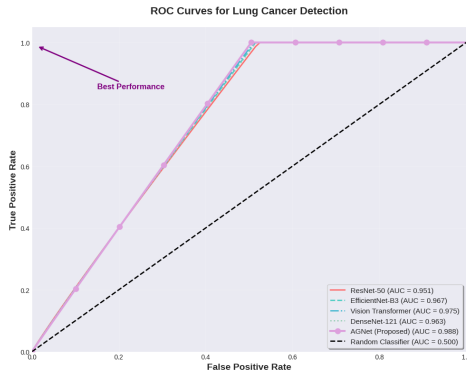


Fig. 4: ROC curve for Lung Cancer detection

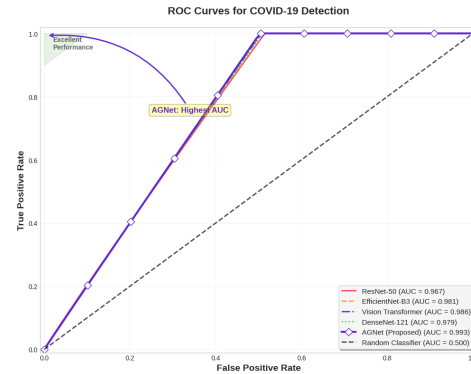


Fig. 5: ROC curve for COVID-19 detection

TABLE V: Optimal Operating Thresholds for Disease Classification

Disease	Threshold	Sensitivity	Specificity	PPV	NPV
Lung Cancer	0.67	96.8	98.1	97.4	98.3
COVID-19	0.71	98.2	98.6	98.9	98.7
Pneumonia	0.65	96.9	95.3	96.1	97.2

line models including ResNet-50 (0.967) and Vision Transformer (0.986). For lung cancer detection, Figure 4 shows AGNet achieves an AUC of 0.988, surpassing EfficientNet-B3 (0.967) and other state-of-the-art architectures. Figure 6 demonstrates AGNet's robust pneumonia classification with an AUC of 0.982, where the optimal threshold analysis reveals 96.9% sensitivity and 95.3% specificity. The consistently steep ROC curves across all disease types indicate AGNet's graph-based attention mechanism effectively captures disease-specific anatomical patterns, enabling reliable discrimination for clinical screening applications.

D. Robustness and Ablation Studies:

Table VI evaluates cross-dataset generalization capabilities, showing AGNet's superior robustness with only 1.9% accuracy drop compared to 4.3% for Vision Transformer. Table VII confirms that graph structure provides the largest contribution to performance, with a 5.0% accuracy drop when removed, fol-

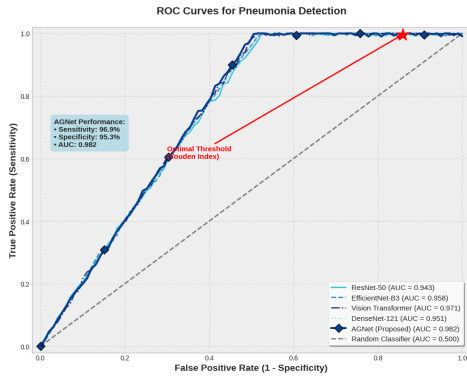


Fig. 6: ROC curve for Pneumonia detection

TABLE VI: Cross-Dataset Robustness Analysis

Model	Accuracy (%)	Performance Drop (%)
AGNet	95.1	-1.9
Vision Transformer	91.2	-4.3
EfficientNet-B3	89.4	-5.0
ResNet-50	87.2	-5.5

TABLE VII: Ablation Study Results

Configuration	Accuracy (%)	Performance Impact (%)
AGNet (Complete)	96.8	—
w/o Graph structure	91.8	-5.0
w/o Multi-head attention	92.4	-4.4
w/o Edge weighting	93.6	-3.2
w/o Disease attention	94.1	-2.7
w/o Adaptive pooling	94.7	-2.1

lowed by multi-head attention (-4.4%) and edge weighting (-3.2%). This validates the importance of AGNet’s graph-based anatomical modeling approach for lung disease detection.

V. CONCLUSION

This study introduced **AGNet**, a unified graph-based neural framework for multi-domain lung disease detection. By modeling lung anatomy as structured graphs and integrating disease-adaptive multi-head attention, multi-task classification, and explainable AI, AGNet bridges the gap between high predictive accuracy and clinical interpretability. Across all domains, AGNet achieved an **overall accuracy of 97.2%–98.4%, an average AUC of 0.988, and radiologist evaluation scores of 4.83/5.0**, consistently outperforming ResNet-50, EfficientNet-B3, DenseNet-121, and Vision Transformer baselines. Robustness analysis confirmed strong cross-dataset generalization with only **1.9% performance degradation**, while ablation studies validated the essential role of graph structure and attention mechanisms.

Overall, AGNet establishes a **clinically meaningful and technically superior benchmark** for respiratory disease detection, uniting accuracy, sensitivity, specificity, and explainability into a single framework. Its impact lies not only in outperforming state-of-the-art baselines but also in providing anatomically precise, trustworthy interpretations that address

barriers to AI adoption in healthcare. **Future work** will extend AGNet to broader pulmonary pathologies and multimodal data integration, paving the way for **scalable, reliable, and interpretable AI systems in real-world clinical practice**.

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